



The Choice of Normalization Directly Affects Feature Selection in Regularized Regression

DSTS's Two-Day Meeting, Autumn 2024

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Problem and Motivation

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Results

- Class balance has a normalization-dependent impact on the model.
- In mixed data, choice of normalization implicitly weighs features' importances.



Figure 1: Joint work with Jonas Wallin

Preliminaries

Motivation and Aims

Results

Experiments

Preliminaries

- Data consists of a **fixed** matrix of features $\mathbf{X} \in \mathbb{R}^{n \times p}$ and a response vector $\mathbf{y} \in \mathbb{R}^n$.
- $\mathbf{y}$ comes from a linear model, that is,

$$y_i = \beta_0^* + \mathbf{x}_i^\top \boldsymbol{\beta}^* + \varepsilon_i \quad \text{for } i \in 1, \dots, n,$$

where $\boldsymbol{\beta}^*$ is the vector of *true* coefficients.

- ε_i is the measurement noise, generated from some random variable.

The Elastic Net

Linear regression plus a combination of the ℓ_1 and ℓ_2 penalties:

$$\boldsymbol{\beta}^* = \arg \min_{\boldsymbol{\beta} \in \mathbb{R}^p} \left(\frac{1}{2} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \underbrace{\lambda_1 \|\boldsymbol{\beta}\|_1}_{\text{lasso}} + \underbrace{\frac{\lambda_2}{2} \|\boldsymbol{\beta}\|_2^2}_{\text{ridge}} \right)$$

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$$\beta^* = \arg \min_{\beta \in \mathbb{R}^p} \left(\underbrace{\frac{1}{2} \|\mathbf{y} - \mathbf{X}\beta\|_2^2}_{\text{least squares}} + \underbrace{\lambda_1 \|\beta\|_1}_{\text{lasso}} + \underbrace{\frac{\lambda_2}{2} \|\beta\|_2^2}_{\text{ridge}} \right)$$

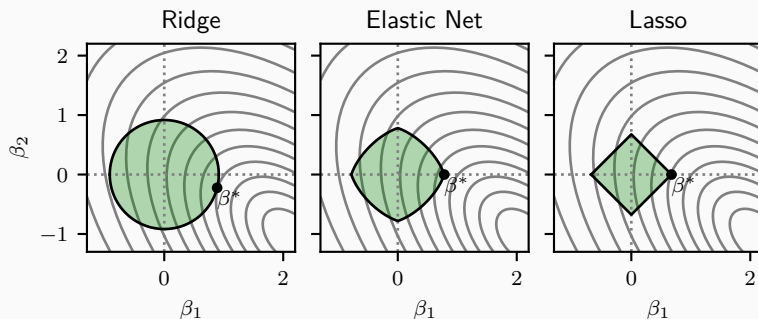


Figure 2: Elastic net is a combination of the lasso and ridge.

Regularization

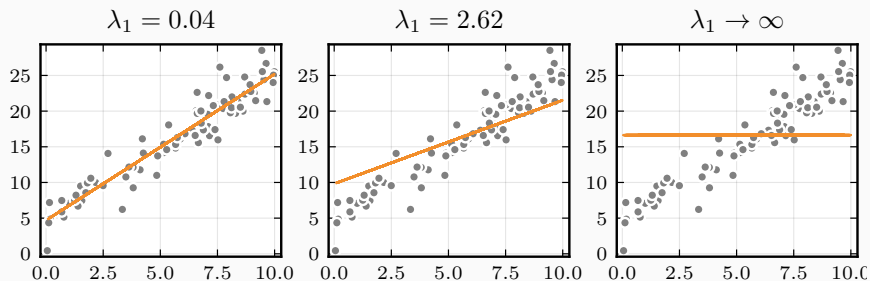


Figure 3: Regularization for a simple linear regression problem

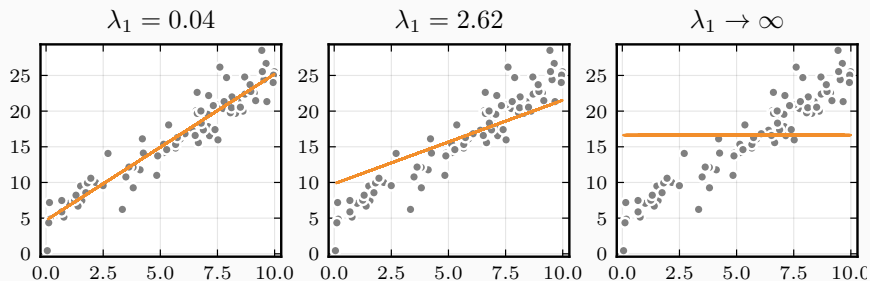


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Why Regularize? (Lasso, Ridge)

- Uniqueness when $p \gg n$
- To overcome overfitting

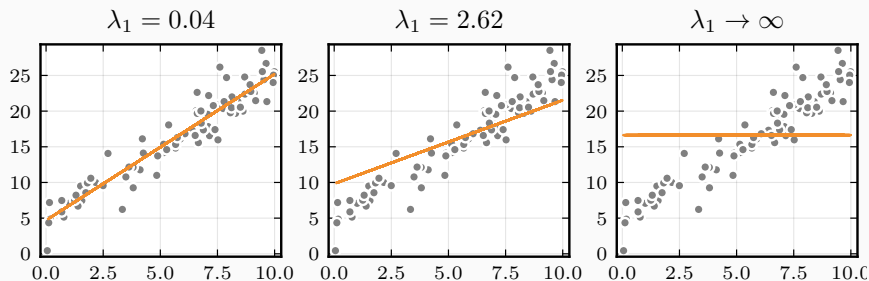


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Why Sparsity? (Lasso)

- Interpretability
- The sparsity bet

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- For each α , solve the elastic net over a sequence of λ : the **elastic net path**.

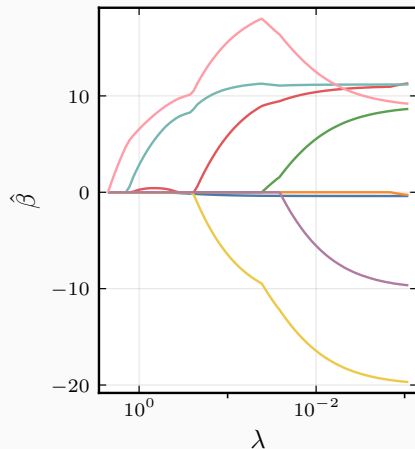


Figure 4: The elastic net path

Sensitivity to Scale

Lasso and ridge penalties are **norms**—feature scales matter!

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Example

Assume

$$\mathbf{X} \sim \text{Normal} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix} \right), \quad \boldsymbol{\beta}^* = \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix}.$$

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Large scale means less penalization because the size of β_j can be smaller for an equivalent effect (on $\mathbf{y}$).

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- After fitting, we transform the coefficients back to their original scale via

$$\hat{\beta}_j = \frac{\hat{\beta}_j^{(n)}}{s_j} \quad \text{for } j = 1, 2, \dots, p,$$

where $\hat{\beta}_j^{(n)}$ is a coefficient from the normalized problem.

Table 1: Common ways to normalize $\mathbf{X}$

Normalization	Centering (c_j)	Scaling (s_j)
Standardization	$\bar{x}_j = \frac{1}{n} \sum_{i=1}^n x_{ij}$	$\sqrt{\frac{1}{n} \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2}$
Min–Max	$\min_i(x_{ij})$	$\max_i(x_{ij}) - \min_i(x_{ij})$
Unit Vector (L2)	0	$\sqrt{\sum_{i=1}^n x_{ij}^2}$
Max–Abs	0	$\max_i(x_{ij})$
Adaptive Lasso	0	β_j^{OLS}

Motivation and Aims

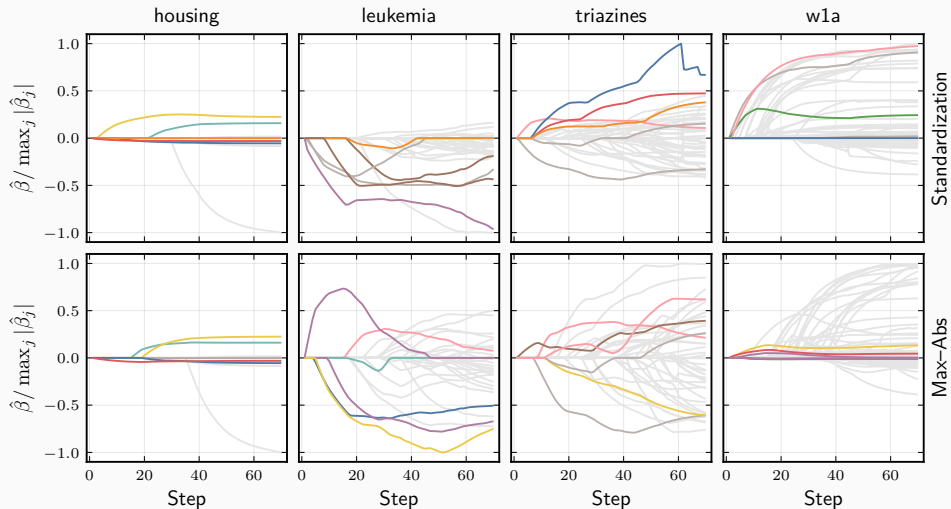


Figure 5: Normalization matters. Lasso paths under two different types of normalization (standardization and max-abs normalization). The union of the first ten features selected in any of the settings are colored.

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Aims

- Binary features, particularly with respect to the **class balance** thereof
- A mix of binary and normally distributed features

Results

Orthogonal Features

There is no explicit solution to the elastic net problem in general (unless $\lambda_1 = 0$).

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But if we assume that the features are orthogonal, that is

$$\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} = \text{diag}(\tilde{\mathbf{x}}_1^\top \tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_p^\top \tilde{\mathbf{x}}_p),$$

then there is:¹:

$$\hat{\beta}_j = \frac{S_{\lambda_1}(\tilde{\mathbf{x}}_j^\top \mathbf{y})}{s_j(\tilde{\mathbf{x}}_j^\top \tilde{\mathbf{x}}_j + \lambda_2)},$$

where

$$S_\lambda(z) = \text{sign}(z) \max(|z| - \lambda, 0).$$

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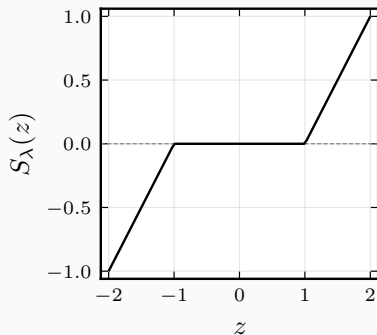


Figure 6: Soft thresholding

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Bias and Variance of the Elastic Net Estimator

The goal is computing the expected value of the elastic net estimator,

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since we treat $\mathbf{X}$ as fixed.

Letting $Z = \tilde{\mathbf{x}}^\top \mathbf{y}$ and assuming that ε_i is i.i.d. Normally-distributed with mean zero and finite variance σ_ε^2 , we have

$$Z \sim \text{Normal}(\mu = \tilde{\mathbf{x}}_j^\top \mathbf{x}_j \beta_j, \sigma^2 = \sigma_\varepsilon^2 \|\tilde{\mathbf{x}}_j\|_2^2).$$

Next, will turn to $\mathbb{E} S_{\lambda_1}(Z)$.

The expected value of the soft-thresholding estimator is

$$\begin{aligned} \mathbb{E} S_{\lambda}(Z) &= \int_{-\infty}^{\infty} S_{\lambda}(z) f_Z(z) \, dz \\ &= \int_{-\infty}^{-\lambda} (z + \lambda) f_Z(z) \, dz \\ &\quad + \int_{\lambda}^{\infty} (z - \lambda) f_Z(z) \, dz. \end{aligned}$$

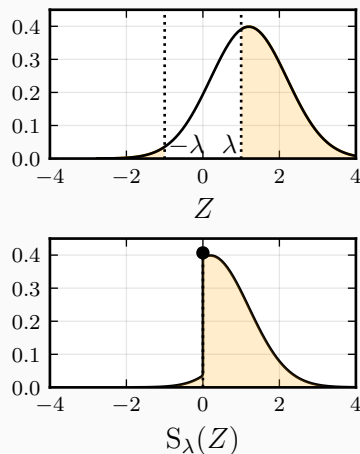


Figure 7: Distributions of Z and its value after soft-thresholding.

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The bias of $\hat{\beta}_j$ is

$$\mathbb{E} \hat{\beta}_j - \beta_j^* = \frac{1}{d_j} \mathbb{E} S_{\lambda}(Z) - \beta_j^*,$$

where $d_j = s_j (\tilde{\mathbf{x}}_j^{\top} \tilde{\mathbf{x}}_j + \lambda_2)$.

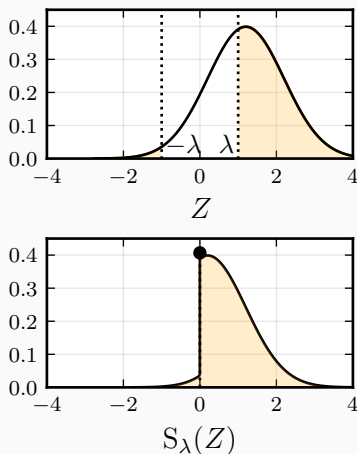


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The variance of the soft-thresholding estimator is

$$\text{Var } S_\lambda(Z) = \int_{-\infty}^{-\lambda} (z + \lambda)^2 f_Z(z) \, dz + \int_{\lambda}^{\infty} (z - \lambda)^2 f_Z(z) \, dz - (\mathbb{E} S_\lambda(Z))^2$$

and consequently the variance of the elastic net estimator is

$$\text{Var } \hat{\beta}_j = \frac{1}{d_j^2} \text{Var } S_\lambda(Z).$$

Recall that

$$Z \sim \text{Normal} \left(\mu = \tilde{\mathbf{x}}_j^\top \mathbf{x}_j \beta_j, \sigma^2 = \sigma_\varepsilon^2 \|\tilde{\mathbf{x}}_j\|_2^2 \right)$$

and assume we have a binary feature $\mathbf{x}_j$, such that $x_{ij} \in \{0, 1\}$. Let $q \in [0, 1]$ be the class balance of this feature, that is: $q = \bar{\mathbf{x}}_j$.

In this case, we observe that

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Binary Features

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And consequently

$$\mu = \frac{\beta_j^* n(q - q^2)}{s_j}, \quad \sigma^2 = \frac{\sigma_\varepsilon^2 n(q - q^2)}{s_j^2}, \quad d_j = \frac{n(q - q^2)}{s_j} + \lambda_2 s_j.$$

Noiseless Case for Binary Features

In the noiseless case, we have

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- Indicates there might be no (simple) s_j that will work for the elastic net.

Probability of Selection

Since $\mathbf{X}$ is fixed and ε is normal, we can compute the probability of selection:

$$\Pr(\hat{\beta}_j \neq 0) = \Phi\left(\frac{\mu - \lambda_1}{\sigma}\right) + \Phi\left(\frac{-\mu - \lambda_1}{\sigma}\right).$$

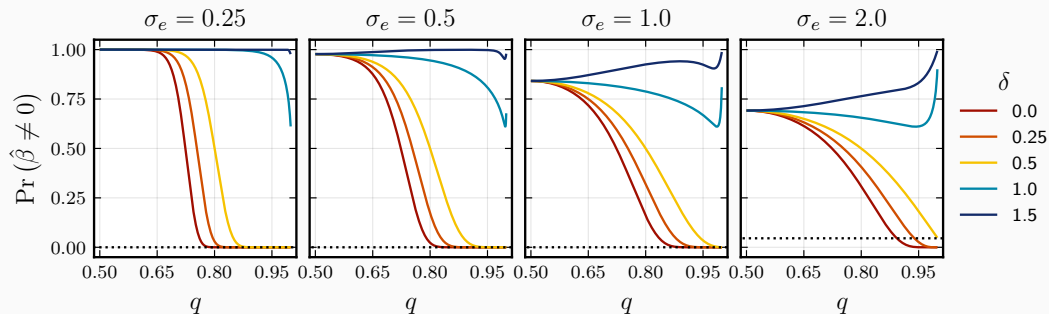


Figure 8: Probability that the elastic net selects a feature across different noise levels (σ_ε), types of normalization (δ), and class balance (q). The dashed line is asymptotic behavior for $\delta = 1/2$. Scaling used is $s_j \propto (q - q^2)^\delta$.

Rare Traits

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Subgroup Data

Results become dependent on data collection.

Collecting more data with different class balances influences the results (since class balances change).

Theorem

If x_j is a binary feature with class balance $q \in (0, 1)$ and $\lambda_1, \lambda_2 \in (0, \infty)$, $\sigma_\varepsilon > 0$, and $s_j = (q - q^2)^\delta$, $\delta \geq 0$, then

$$\lim_{q \rightarrow 1^+} \mathbb{E} \hat{\beta}_j = \begin{cases} 0 & \text{if } 0 \leq \delta < \frac{1}{2}, \\ \frac{2n\beta_j^*}{n+\lambda_2} \Phi\left(-\frac{\lambda_1}{\sigma_\varepsilon\sqrt{n}}\right) & \text{if } \delta = \frac{1}{2}, \\ \beta_j^* & \text{if } \delta \geq \frac{1}{2}. \end{cases}$$

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and

$$\lim_{q \rightarrow 1^+} \text{Var} \hat{\beta}_j = \begin{cases} 0 & \text{if } 0 \leq \delta < \frac{1}{2}, \\ \infty & \text{if } \delta \geq \frac{1}{2}. \end{cases}$$

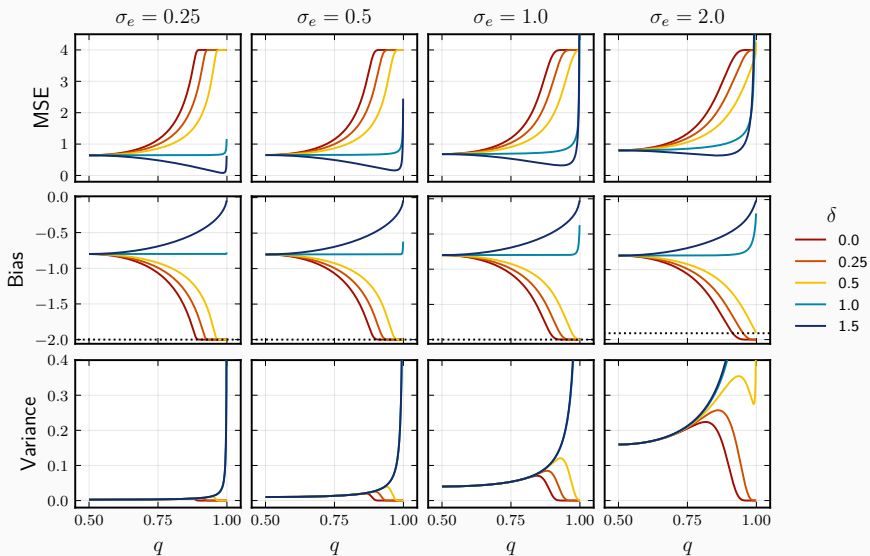
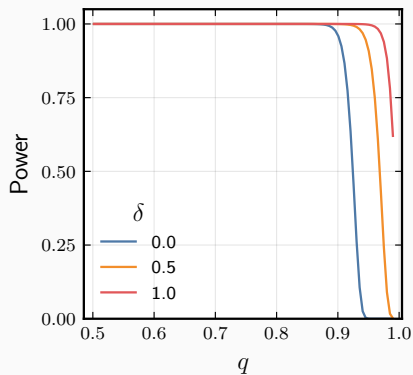
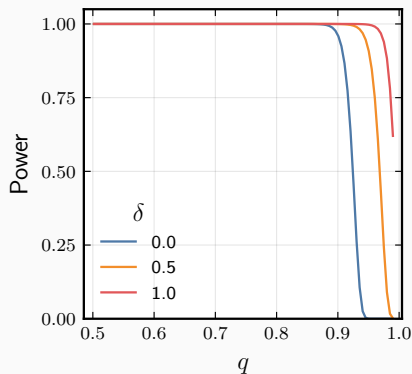


Figure 9: A bias variance tradeoff. Bias, variance, and mean-squared error for a one-dimensional lasso problem. Theoretical result for orthogonal features. Dotted line is asymptotic result or $\delta = 1/2$. Scaling used is $s_j \propto (q - q^2)^\delta$.

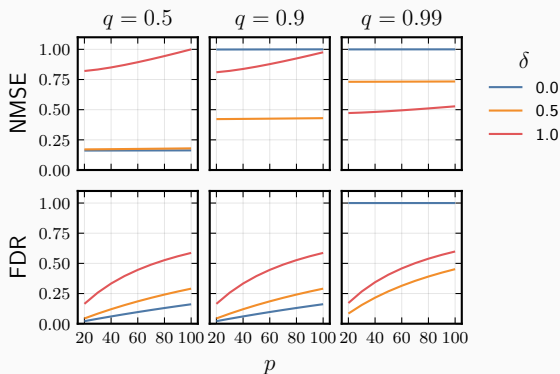


(a) Power in the sense of detecting all the true signals. Constant p .

Figure 10: Multiple features: 10 true signals and varying q and p . Mean squared error (MSE), false discovery rate (FDR), and power



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(b) False discovery rate (FDR) and normalized mean-squared error (NMSE).

Figure 10: Multiple features: 10 true signals and varying q and p . Mean squared error (MSE), false discovery rate (FDR), and power

So far: all binary features. What about mixing binary and continuous (normal) features?

How to put binary features and normal features on the “same” scale?

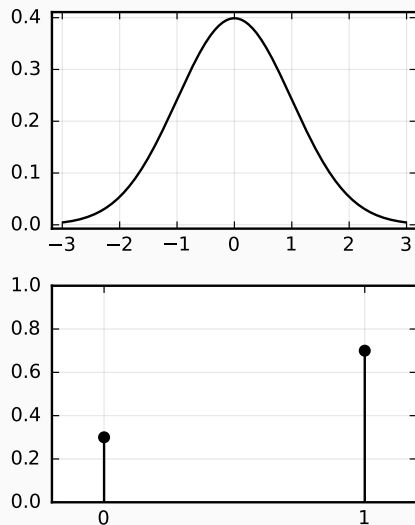


Figure 11: How do we match these?

Our Definition of Comparability

The effects of a binary feature and a normally distributed feature are **comparable** if a flip in the binary feature has the same effect as a two-standard deviation change in the normal feature (Gelman 2008).

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Examples

Assume entries in x_1 are binary and x_2 come from a random variable X_2 . The effects are comparable in the following cases:

- $X_2 \sim \text{Normal}(\mu, 1/2)$, $\beta_1^* = 1$, and $\beta_2^* = 1$.
- $X_2 \sim \text{Normal}(\mu, 2)$, $\beta_1^* = 1$, and $\beta_2^* = 0.25$.

Choice of Scaling in Mixed Data

For the two-standard deviation notion of comparability to hold, we need to modify our scaling factor s_j .

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As before, we assume that x_1 is binary and $X_2 \sim \text{Normal}(\mu, 1/2)$, $\beta_1^* = \beta_2^* = 1$ so that they have *comparable* effects. Also assume we standardize x_2 .

We want $\hat{\beta}_1 = \hat{\beta}_2$. That is,

$$\underbrace{\frac{S_{\lambda_1} \left(\frac{n(q-q^2)}{s_j} \right)}{s_1 \left(\frac{n(q-q^2)}{s_1^2} + \lambda_2 \right)}}_{\hat{\beta}_1} = \underbrace{\frac{S_{\lambda_1} \left(\frac{n}{2} \right)}{\frac{1}{2} (n + \lambda_2)}}_{\hat{\beta}_2}.$$

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The choice $s_1 = (2(q - q^2))^\delta$ works when classes are balanced ($q = 0.5$). But no clear choice for the elastic net case.

Experiments

Binary Features (Decreasing q)

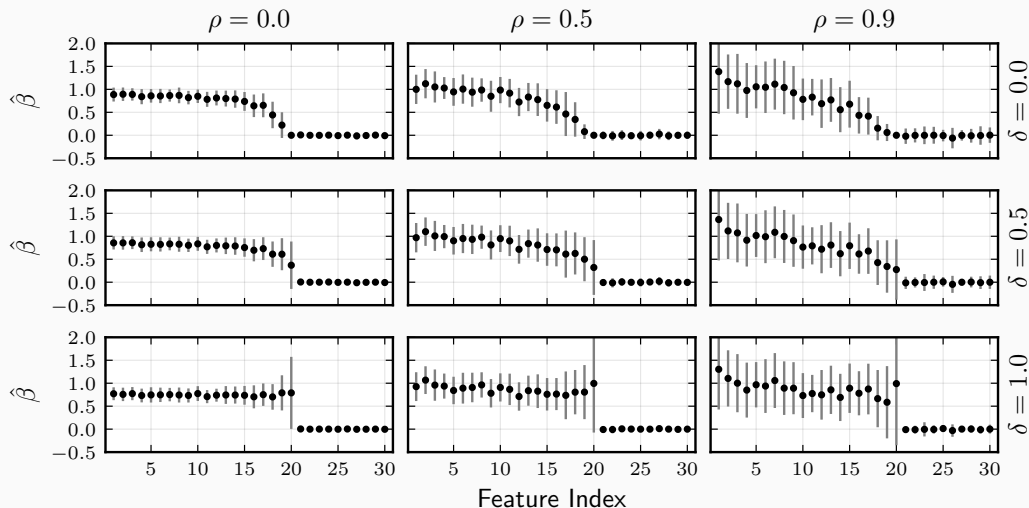


Figure 12: Lasso estimates for first 30 coefficients. First 20 features are true signals with a geometrically decreasing class balance from 0.5 to 0.99. ρ is a measure of autocorrelation.

In the middle

Binary Features (Signal-to-Noise Ratio)

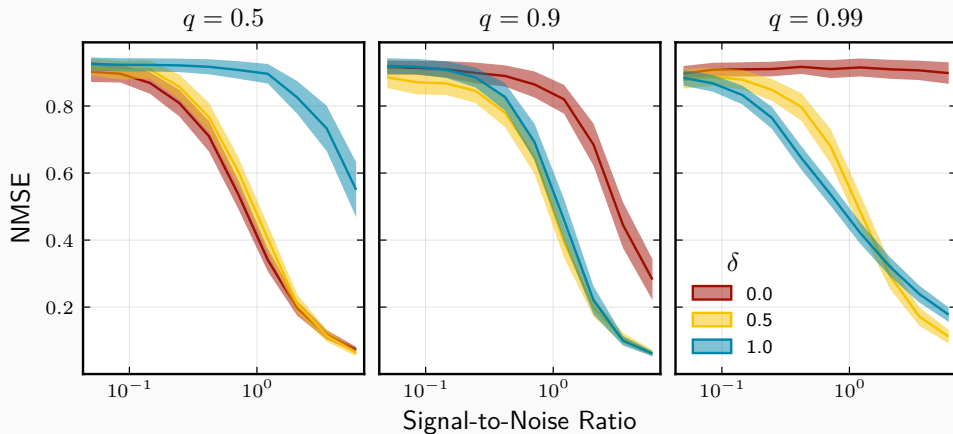


Figure 13: Normalized mean-squared test set error (NMSE).

Mixed Data

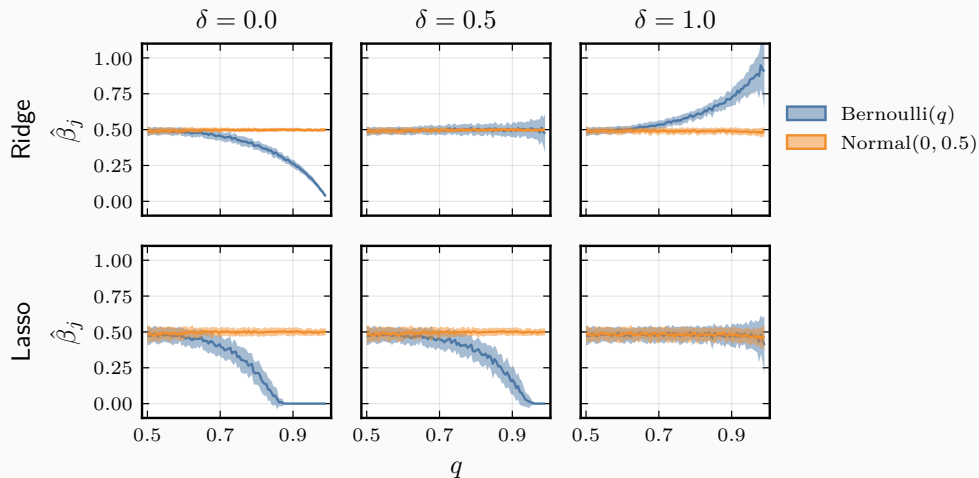


Figure 14: Comparison between lasso and ridge estimators for features generated to resemble features from various distributions.

Hyperparameter Optimization

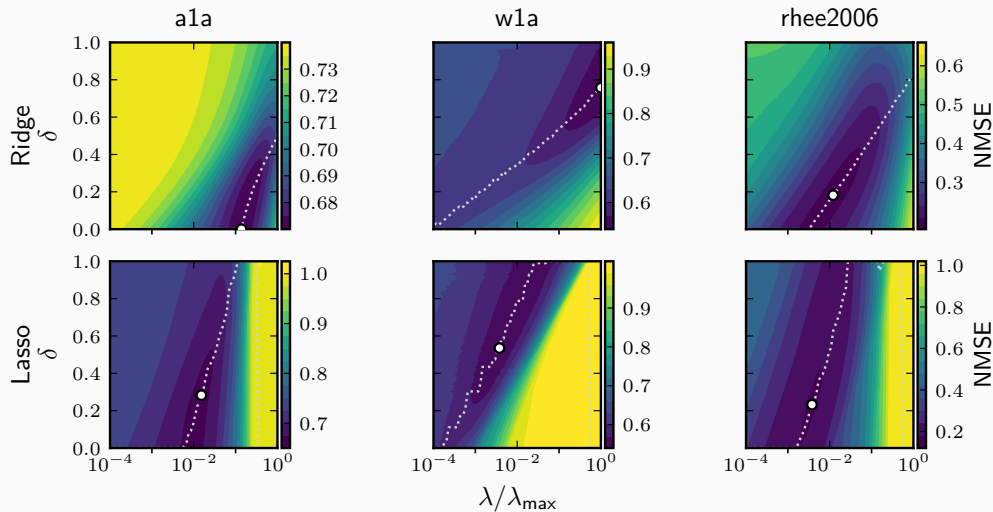


Figure 15: Contour plots of hold-out (validation set) error across a grid of δ and λ values for the lasso and ridge.

Conclusions

- Class balance plays a crucial role when using regularized regression on binary data.
- As far as we know the first paper to investigate the interplay between normalization and regularization
- New scaling approach to deal with class-imbalanced binary features
- Discussion and suggestions for dealing with mixed data

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Limitations

- So far only theoretical results for limited cases:
 - Fixed data ($\mathbf{X}$), normal noise
 - Orthogonal features
 - Normal and binary features

Thank you!

-  El Ghaoui, Laurent, Vivian Viallon, and Tarek Rabbani (Sept. 21, 2010). *Safe Feature Elimination in Sparse Supervised Learning*. Technical report UCB/EECS-2010-126. Berkeley: EECS Department, University of California. URL: <http://www.eecs.berkeley.edu/Pubs/TechRpts/2010/EECS-2010-126.html>.
-  Gelman, Andrew (July 10, 2008). "Scaling Regression Inputs by Dividing by Two Standard Deviations". In: *Statistics in Medicine* 27.15, pp. 2865–2873. ISSN: 02776715, 10970258. DOI: [10.1002/sim.3107](https://doi.org/10.1002/sim.3107). URL: <https://onlinelibrary.wiley.com/doi/10.1002/sim.3107> (visited on 09/27/2023).
-  Tibshirani, Robert et al. (Mar. 2012). "Strong Rules for Discarding Predictors in Lasso-Type Problems". In: *Journal of the Royal Statistical Society. Series B: Statistical Methodology* 74.2, pp. 245–266. ISSN: 1369-7412. DOI: [10/c4bb85](https://doi.org/10.1016/j.jrssb.2012.03.001). URL: <https://iths.pure.elsevier.com/en/publications/strong-rules-for-discarding-predictors-in-lasso-type-problems> (visited on 03/16/2018).

-  Zou, Hui and Trevor Hastie (2005). “Regularization and Variable Selection via the Elastic Net”. In: *Journal of the Royal Statistical Society. Series B (Statistical Methodology)* 67.2, pp. 301–320. ISSN: 1369-7412. URL: www.jstor.org/stable/3647580 (visited on 03/12/2018).

Extras

Max-Abs Scaling of Continuous Features

- Min-max normalization is sometimes used in continuous data
- Very sensitive to outliers
- But also depend on sample size!
- In other words, results in model validation with varying sample sizes can yield very strange results.

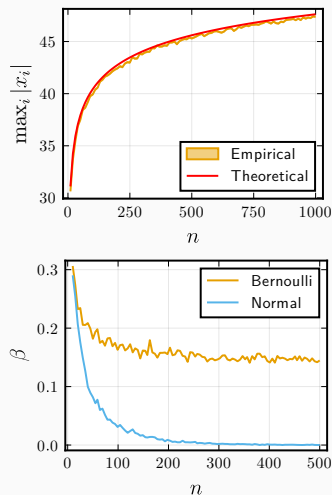


Figure 16: Effects of maximum absolute value scaling.

Hyperparameter Optimization (Support and NMSE)

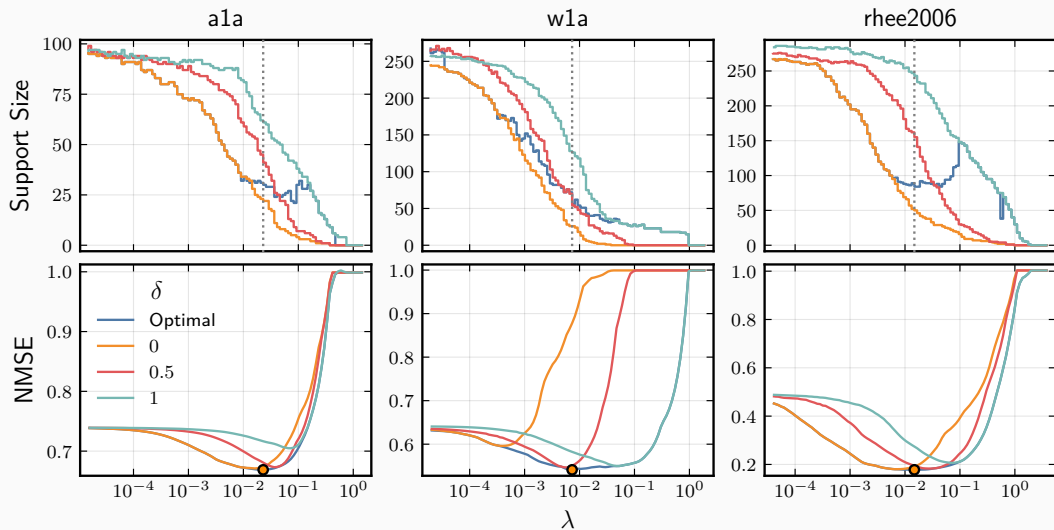


Figure 17: Support and NMSE of the lasso for different values of δ and λ .

Background on the Elastic Net

- Proposed by Zou and Hastie (2005).

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- Lasso (ℓ_1) part:
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- Ridge (ℓ_2) part
 - Mitigates lasso issue in correlated data
 - Better predictive performance when true signal is non-sparse